Technical Report: Scenario Generation for Context-Aware Battery Thermal Management: Mode-Switching MPC with SVM-HMM Driving-Context Forecasting

Esra Hicret Uzer [1] and Tomáš Haniš[2]

***Abstract*— This technical report presents the modeling framework, system identification methodology, and simulation setup developed for a context-aware predictive battery thermal management system (BTMS) for battery electric vehicles. The report documents the electro-thermal battery model, heat generation formulation, battery aging and state-of-health estimation framework, and the control-oriented state-space representation employed for prediction and control development. In addition, a comprehensive scenario generation methodology is introduced to capture the effects of environmental conditions, vehicle dynamics, traffic conditions, and driver behavior on thermal loading. The generated operating scenarios are intended to support the development and validation of predictive thermal management strategies under realistic driving conditions. This document serves as a companion reference to the accompanying conference publication by providing detailed model derivations, parameter definitions, assumptions, and simulation settings required for reproducibility and future implementation within a realistic digital twin environment.**

I. INTRODUCTION

Lithium-ion batteries must operate within a strict and narrow temperature range (20°C–40°C) while maintaining minimal thermal gradients [1], [2]. Exceeding the optimal upper temperature limits accelerates the continuous thickening of the solid electrolyte interphase (SEI) layer on the anode [3], [4]. This degradation process irreversibly consumes active cyclic lithium and increases internal charge-transfer resistance, ultimately leading to severe capacity fade [3]. Conversely, operating at low temperatures severely reduces lithium-ion mobility and diffusion rates. Under high charging currents in cold environments, this sluggish diffusion triggers lithium plating—a condition where metallic lithium deposits on the anode surface, creating dendrites that can pierce the separator and cause catastrophic internal short circuits [4]

Traditional thermal monitoring typically relies on surface-mounted temperature sensors; however, large-format lithium-ion cells exhibit significant temperature gradients between the core and the surface due to substantial thermal mass and poor cross-plane thermal conductivity [1]. Research indicates that internal core temperatures can drastically diverge from surface measurements, particularly during high-rate transient discharge and charge events [1], [2]. These extreme spatial temperature gradients induce non-uniform current distribution and localized electrochemical stress, further accelerating degradation mechanisms like localized lithium plating [5]. Therefore, it is critical to not only keep both the surface and core temperatures within the optimal bounds but also strictly limit the spatial gradient between them (often restricted with less than 5°C) to prevent asymmetric aging, material cracking, and severe safety risks [1], [2].

To maintain these strict thermal requirements, accurate temperature prediction is essential. Historically, control frameworks have evaluated battery thermal behavior using standardized, static drive cycle tests (WLTP, NEDC etc.) [6]. However, these standard static cycles are highly idealized and insufficient for accurate thermal prediction [6]. Real-world vehicle thermal loading and energy consumption are stochastic and heavily influenced by a combination of diverse dynamic parameters, including ambient weather conditions, real-time traffic dynamics, road topology, auxiliary loads, and driver behavior [6], [7]. Evaluating predictive systems exclusively on standard cycles fails to capture these complex interactions, leading to compromised pack longevity. Incorporating these comprehensive contextual parameters is vital to capturing accurate predictive thermal modeling under realistic conditions [6].

To bring high-level stochastic driving contexts and real-time optimal thermal control together, this technical report documents the mathematical foundations of the heat generation mechanism and real-world driving scenarios with a variety of operational conditions. This document serves as the formal mathematical reference and implementation manual for the accompanying predictive digital-twin framework.

II. SYSTEM MODEL

A. Cell to Pack Scaling

To maintain consistency between cell-level electrochemical states and vehicle-level power demands, all parameters and inputs are mapped across the pack topology. The high-voltage battery pack is arranged in an N_sSN_pP matrix configuration ($N_s = 108$, $N_p = 2$). The corresponding

This research was supported by the Grant Agency of the Czech Technical University in Prague, grant No. SGS25/145/OHK3/3T/13

[1]Esra Hicret Uzer is with the Faculty of Electrical Engineering, Czech Technical University in Prague, Prague 120 00, Czech Republic uzeresra@fel.cvut.cz

[2]Tomáš Haniš is with the Faculty of Electrical Engineering, Czech Technical University in Prague, Prague 120 00, Czech Republic tomas.hanis@cvut.cz

NOMENCLATURE

A	Frontal cross-sectional area $[\mathrm{m}^2]$
a	Instantaneous vehicle acceleration $[\mathrm{m/s}^2]$
C_c	Lumped thermal capacitance of cell core $[\mathrm{J/K}]$
C_d	Dimensionless aerodynamic drag coefficient $[-]$
C_s	Lumped thermal capacitance of casing $[\mathrm{J/K}]$
C_{bat}	Nominal individual cell charge capacity $[\mathrm{A \cdot s}]$
C_1, C_2	Polarization capacitances per cell [F]
E_{batt}	Aggregate electrical pack capacity [kWh]
F_{tr}	Instantaneous tire tractive force demand [N]
f_r	Dimensionless rolling resistance coefficient $[-]$
g	Gravitational acceleration constant $[\mathrm{m/s}^2]$
I_{cell}	Cell current, positive discharge [A]
I_{batt}	Aggregate pack terminal current [A]
$I_{1\mathrm{C}}$	1C reference continuous current benchmark [A]
M	Base vehicle curb mass [kg]
m	Total vehicle mass ($M + m_{\mathrm{load}}$) [kg]
m_{load}	Variable passenger and cargo payload [kg]
N_p	Parallel cell branches per series string $[-]$
N_s	Cells connected in electrical series $[-]$
P_{aux}	Total auxiliary electrical load [W]
P_{elec}	Net pack terminal electrical power load [W]
$Q_{\mathrm{gen,cell}}$	Cell internal heat generation rate [W]
Q_{pack}	Aggregate pack thermal generation rate [W]
$\bar{Q}_{\mathrm{pack}}$	Window-averaged pack thermal rate [W]
Q_{nom}	Nominal cell charge capacity benchmark $[\mathrm{A \cdot h}]$
R_c	Core-to-surface conductive resistance $[\mathrm{K/W}]$
R_g	Universal gas constant $[\mathrm{J/(mol \cdot K)}]$
R_s	High-frequency ohmic resistance per cell $[\Omega]$
$R_{s,0}$	Pristine cell baseline ohmic resistance $[\Omega]$
$R_{s,\mathrm{pack}}$	Aggregate battery pack series resistance $[\Omega]$
R_u	Casing-to-coolant convective resistance $[\mathrm{K/W}]$
R_1, R_2	Polarization overpotential resistances $[\Omega]$
SOC	Battery cell state of charge $[-]$
SOH	Battery cell state of health $[-]$
T_c	Battery cell core temperature [K]
T_f	Coolant fluid boundary temperature [K]
T_s	Surface casing temperature [K]
T_w	Moving-average evaluation window [s]
U_{batt}	Nominal pack bus voltage [V]
V_{OCV}	Individual cell open-circuit voltage [V]
$V_{\mathrm{eq,pack}}$	Equivalent aggregate open-circuit voltage [V]
V_{pack}	Battery pack terminal voltage under load [V]
$V_{RC,1}, V_{RC,2}$	Polarization branch overvoltages [V]
V_T	Cell terminal voltage under load [V]
v	Longitudinal vehicle velocity $[\mathrm{m/s}]$
v_{cut}	Cut-off speed for regenerative braking $[\mathrm{m/s}]$
v_{limit}	Route segment statutory speed limit $[\mathrm{km/h}]$
w	Headwind longitudinal velocity $[\mathrm{m/s}]$

Greek symbols

α_{regen}	Regenerative brake blending fraction $[-]$
δ	Rotational inertia mass equivalent factor $[-]$
η_{pt}	Driveline mechanical-traction efficiency $[-]$
η_{regen}	Dynamic regenerative capture efficiency $[-]$
θ	Road inclination grade angle $[\mathrm{rad}, {}^\circ]$
λ_{brk}	Regenerative braking cut-off slope $[\mathrm{s/m}]$
μ	Road-tire surface friction coefficient $[-]$
ρ	Ambient air mass density $[\mathrm{kg/m}^3]$
$\frac{\partial V_{OCV}}{\partial T_c}$	Entropic potential coefficient $[\mathrm{V/K}]$

Subscripts

amb	Ambient environmental condition
aux	Auxiliary vehicle electrical systems
batt	High-voltage battery pack aggregate
brk	Mechanical or regenerative braking system
c	Core region of the battery cell
cell	Single electrochemical unit cell
elec	Net electrical traction domain
pt	Powertrain and inverter components
regen	Dynamic regenerative energy capture
s	Outer surface casing of the cell
tr	Wheel-road interface tractive tract
u	Convective fluid boundary interface

scaling transformations are defined as:

$$I_{\text{cell}} = \frac{I_{\text{batt}}}{N_p}, \tag{1}$$

$$V_{\text{pack}} = N_s V_T, \tag{2}$$

$$Q_{\text{pack}} = N_s N_p Q_{\text{gen,cell}}, \tag{3}$$

$$R_{s,\text{pack}} = \frac{N_s}{N_p} R_s, \tag{4}$$

$$C_{i,\text{pack}} = \frac{N_p}{N_s} C_i, \quad i \in \{1,2\}, \tag{5}$$

$$C_{\text{bat,pack}} = N_p C_{\text{bat}}. \tag{6}$$

B. Coupled Electro-Thermal Model

The cell-level electrical behavior is governed by a second-order Equivalent Circuit Model (ECM) comprising a static nonlinear open-circuit voltage source $V_{OCV}(\text{SOC})$, an ohmic series resistance R_s, and two dynamic polarization RC parallel pairs (R_1–C_1, R_2–C_2). The governing state equations for cell state-of-charge and polarization overvoltages are:

$$\dot{\text{SOC}} = -\frac{I_{\text{cell}}}{C_{\text{bat}}}, \tag{7}$$

$$\dot{V}_{RC,1} = -\frac{V_{RC,1}}{R_1 C_1} + \frac{I_{\text{cell}}}{C_1}, \tag{8}$$

$$\dot{V}_{RC,2} = -\frac{V_{RC,2}}{R_2 C_2} + \frac{I_{\text{cell}}}{C_2}, \tag{9}$$

subject to the observable terminal cell voltage output equation:

$$V_T = V_{OCV}(\text{SOC}) - I_{\text{cell}} R_s - V_{RC,1} - V_{RC,2}. \tag{10}$$

Internal cell heat generation $Q_{\text{gen,cell}}$ accounts for irreversible Joule dissipation, polarization overpotential losses, and reversible entropic heat generation [9]:

$$\begin{aligned} Q_{\text{gen,cell}} = {} & I_{\text{cell}}^2 R_s(T_c, L_{\text{sei}}) + I_{\text{cell}} V_{RC,1} + I_{\text{cell}} V_{RC,2} \\ & - I_{\text{cell}} T_c^{\text{abs}} \frac{\partial V_{OCV}(\text{SOC})}{\partial T_c}, \end{aligned} \tag{11}$$

where T_c^{abs} denotes the absolute core temperature in Kelvin, ensuring consistent thermodynamic evaluation of reversible heat dissipation.

Thermal evolution inside the cell is captured via a two-state core-surface formulation [10]. Conductive transport occurs internally across thermal resistance R_c, while convective boundary heat removal operates over surface resistance R_u into the controllable fluid boundary T_f:

$$C_c \dot{T}_c = Q_{\text{gen,cell}} - \frac{T_c - T_s}{R_c}, \tag{12}$$

$$C_s \dot{T}_s = \frac{T_c - T_s}{R_c} - \frac{T_s - T_f}{R_u}. \tag{13}$$

C. State of Health and SEI Growth Over Time

Degradation dynamics operate over operational timescales orders of magnitude slower than thermal and electrical transients ($\tau_{\text{aging}} \gg 10^5\,\text{s}$ vs. $\tau_{\text{thermal}} \sim 10^2\,\text{s}$). Consequently, the solvent-reduction solid electrolyte interphase (SEI) layer propagation is decoupled from the fast predictive control horizon and integrated as a quasi-static observer state. The film thickness growth rate accounts for the pristine passivated native film thickness $L_{\text{sei},0}$:

$$\frac{dL_{\text{sei}}}{dt} = \frac{k_{\text{sei}}}{2\big(L_{\text{sei},0} + L_{\text{sei}}(t)\big)} \exp\left(-\frac{E_{a,\text{sei}}}{R_g T_c^{\text{abs}}}\right) f_{\text{sei}}(\text{SOC}), \tag{14}$$

subject to $L_{\text{sei}}(0) = 0$, with baseline thickness $L_{\text{sei},0} = 10 \times 10^{-9}\,\text{m}$. The non-dimensional state-of-charge acceleration function is structured empirically as:

$$f_{\text{sei}}(\text{SOC}) = \exp(\beta_{\text{sei}} \cdot \text{SOC}). \tag{15}$$

Capacity fade accounts for both continuous calendar degradation and dynamic cycling-induced mechanical stress:

$$\dot{Q}_{\text{fade}} = \gamma_{\text{sei}} \frac{dL_{\text{sei}}}{dt} \left(1 + \xi_{\text{cyc}} \frac{|I_{\text{cell}}|}{I_{1\text{C}}}\right), \tag{16}$$

where γ_{sei} links volumetric film propagation to active lithium loss, and ξ_{cyc} scales current-induced crack propagation. The resulting SEI thickness expands internal resistance across long cycles, updating the electro-thermal model parameters:

$$R_s(t, T_c, L_{\text{sei}}) = R_{s,0}(T_c) + k_R\big(L_{\text{sei},0} + L_{\text{sei}}(t)\big). \tag{17}$$

D. Control-Oriented State-Space Model

From a control synthesis perspective, battery current I_{cell} is an exogenous disturbance imposed by vehicle motion, while the coolant fluid temperature T_f serves as the manipulated control input. Defining the state vector $\mathbf{x} = [\text{SOC}, V_{RC,1}, V_{RC,2}, T_c, T_s]^T \in \mathbb{R}^5$, the scalar control input $u = T_f \in \mathbb{R}$, the exogenous disturbance input $w = I_{\text{cell}} \in \mathbb{R}$, and the output vector $\mathbf{y} = [V_T, T_s]^T \in \mathbb{R}^2$, the open-loop dynamics are formulated as:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}_u u + \mathbf{B}_w w + \mathbf{\Phi}(\mathbf{x}, w), \tag{18}$$

$$\mathbf{y} = \mathbf{h}(\mathbf{x}, w). \tag{19}$$

The continuous-time system matrix $\mathbf{A} \in \mathbb{R}^{5\times5}$ is populated as:

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & a_{22} & 0 & 0 & 0 \\ 0 & 0 & a_{33} & 0 & 0 \\ 0 & 0 & 0 & a_{44} & -a_{44} \\ 0 & 0 & 0 & a_{54} & a_{55} \end{bmatrix}, \tag{20}$$

with the individual non-zero entries defined as:

$$a_{22} = -\frac{1}{R_1 C_1}, \quad a_{33} = -\frac{1}{R_2 C_2}, \tag{21}$$

$$a_{44} = -\frac{1}{R_c C_c}, \quad a_{54} = \frac{1}{R_c C_s}, \tag{22}$$

$$a_{55} = -\left(\frac{1}{R_c C_s} + \frac{1}{R_u C_s}\right). \tag{23}$$

The control input distribution matrix $\mathbf{B}_u \in \mathbb{R}^{5\times1}$ and exogenous disturbance input distribution matrix $\mathbf{B}_w \in \mathbb{R}^{5\times1}$

are defined as:

$$\mathbf{B}_u = \begin{bmatrix} 0 & 0 & 0 & 0 & \dfrac{1}{R_u C_s} \end{bmatrix}^T, \tag{24}$$

$$\mathbf{B}_w = \begin{bmatrix} -\dfrac{1}{C_{\text{bat}}} & \dfrac{1}{C_1} & \dfrac{1}{C_2} & 0 & 0 \end{bmatrix}^T. \tag{25}$$

The nonlinear thermal generation coupling vector $\mathbf{\Phi}(\mathbf{x}, w) \in \mathbb{R}^{5\times 1}$ isolates internal heat injection:

$$\mathbf{\Phi}(\mathbf{x}, w) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \dfrac{1}{C_c} Q_{\text{gen,cell}}(\mathbf{x}, w) \\ 0 \end{bmatrix}, \tag{26}$$

with $Q_{\text{gen,cell}}(\mathbf{x}, w)$ defined via (11). The nonlinear measurement equation $\mathbf{h}(\mathbf{x}, w)$ is:

$$\mathbf{h}(\mathbf{x}, w) = \begin{bmatrix} V_{OCV}(\text{SOC}) - wR_s(\mathbf{x}) - V_{RC,1} - V_{RC,2} \\ T_s \end{bmatrix}. \tag{27}$$

III. SCENARIO GENERATION AND SIMULATION SETUP

A. Thermal Load Generation

Vehicle dynamics drive the instantaneous electrical loading placed on the high-voltage pack. The longitudinal tractive force F_{tr} demanded at the tire contact patch balances aerodynamic drag, rolling resistance, gravitational incline forces, and translational/rotational inertia:

$$\begin{aligned} F_{tr} = f_r mg\cos(\theta) + \frac{1}{2}\rho A C_d (v - w)^2 \\ + mg\sin(\theta) + \delta ma. \end{aligned} \tag{28}$$

Accounting for auxiliary power demands:

$$P_{\text{aux}} = P_{\text{lights}} + P_{\text{climate}} + P_{\text{wipers}} + P_{\text{radio\&nav}} + 30\,\text{W}, \tag{29}$$

the net electrical power demanded at the battery terminals P_{elec} is formulated by separating positive traction from regenerative braking phases:

$$P_{\text{elec}}(t) = \begin{cases} \dfrac{F_{tr}(t)v(t)}{\eta_{\text{pt}}} + P_{\text{aux}}(t), & F_{tr}(t) \geq 0, \\ \alpha_{\text{regen}}(v)\,\eta_{\text{regen}} F_{tr}(t)v(t) + P_{\text{aux}}(t), & F_{tr}(t) < 0, \end{cases} \tag{30}$$

When $F_{tr}(t) < 0$, the tractive power product $F_{tr}(t)v(t) < 0$, ensuring charging electrical power flows into the pack. The velocity-dependent fraction $\alpha_{\text{regen}}(v) \in [0, 1]$ smoothly cuts off regenerative capture at low speeds:

$$\alpha_{\text{regen}}(v) = \frac{1}{1 + \exp\big(-\lambda_{\text{brk}}(v - v_{\text{cut}})\big)}, \tag{31}$$

with $v_{\text{cut}} = 2.5\,\text{m/s}$ and $\lambda_{\text{brk}} = 2.0\,\text{s/m}$.

The demanded pack terminal current I_{batt} is obtained via power conservation across the equivalent source [11] :

$$P_{\text{elec}} = \Big(V_{\text{eq,pack}} - I_{\text{batt}} R_{s,\text{pack}}\Big) I_{\text{batt}}, \tag{32}$$

where $V_{\text{eq,pack}} = N_s\big(V_{OCV}(\text{SOC}) - V_{RC,1} - V_{RC,2}\big)$.

B. Telemetry & Contextual Feature Generation

To train and validate predictive controllers, a dataset of $N = 1296$ distinct stochastic drive profiles is synthesized from established vehicular baseline data [8], [11], [13]. Driving profiles span variations across four contextual domains structured as a full factorial grid:

$$N = N_{T_{\text{amb}}} \times N_\mu \times N_\theta \times N_{m_{\text{load}}} \times N_{\text{route}} \times N_{\kappa_{\text{agg}}} = 1296, \tag{33}$$

where the parameter levels comprise: 4 ambient temperatures ($N_{T_{\text{amb}}} = 4$), 3 friction levels ($N_\mu = 3$), 3 road grades ($N_\theta = 3$), 3 payload levels ($N_{m_{\text{load}}} = 3$), 4 route types ($N_{\text{route}} = 4$), and 3 driver urgency levels ($N_{\kappa_{\text{agg}}} = 3$), yielding $4 \times 3 \times 3 \times 3 \times 4 \times 3 = 1296$ distinct scenarios.

Driver urgency $\kappa_{\text{agg}} \in [0.2, 1.0]$ parameterizes driving aggressiveness by modulating longitudinal acceleration envelope limits and velocity tracking dynamics:

$$a_{\max}(\kappa_{\text{agg}}) = a_{\text{base}} \cdot \big(0.5 + 0.5\kappa_{\text{agg}}\big). \tag{34}$$

The discrete-time feature state $\mathbf{z}(k) \in \mathbb{R}^4$ monitored by the prediction layer is formulated as:

$$\mathbf{z}(k) = \begin{bmatrix} T_c(k) & P_{\text{elec}}(k) & I_{\text{batt}}(k) & a(k) \end{bmatrix}^T. \tag{35}$$

TABLE I
MODEL PARAMETERS, VEHICLE SPECIFICATIONS, AND SCENARIO GRID

Parameter	Value
Cell nominal capacity	78
Baseline ohmic resistance	0.92×10^{-3}
Polarization resistance 1	0.65×10^{-3}
Polarization resistance 2	0.85×10^{-3}
Polarization capacitance 1	2450
Polarization capacitance 2	18 200
Cell core heat capacity	735.0
Cell surface heat capacity	231.0
Vehicle curb mass	1805
Frontal cross-sectional area	2.36
Air resistance coefficient	0.27
Rolling resistance coefficient	0.012
Rotating mass coefficient	1.05
Total pack electrical energy	$\sim$ 58 (NCM 712)
Nominal pack bus voltage	400 (108S2P)
Ambient temperature	{0, 5, 25, 45}
Road surface friction	{0.35, 0.65, 0.95}
Road gradient	{−5, 0, +8}
Route speed limit	{35, 70, 100, 150}
Driver urgency factor	{0.25, 0.65, 1.00}
*Parameters synthesized from [4],[10]–[12].	

IV. CONCLUSIONS

This report presented the mathematical models, assumptions, parameter definitions, and scenario-generation methodology used in the development of a context-aware predictive battery thermal management framework for battery electric vehicles. The document described a coupled electro-thermal battery model, a control-oriented thermal representation structured explicitly into manipulated and disturbance input paths, and a decoupled slow-timescale battery aging

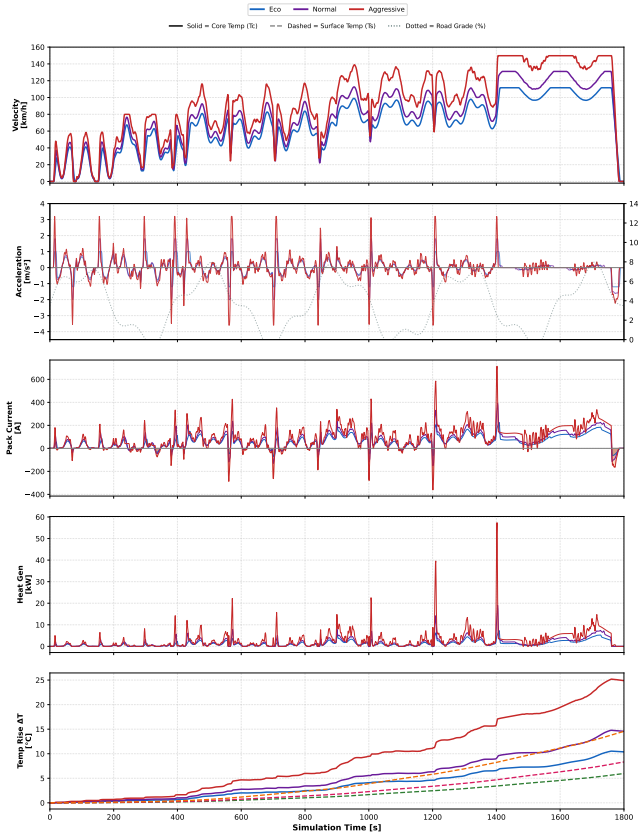


Fig. 1. Simulated vehicle states and battery thermal response under varying operational and road conditions. The driving scenario incorporates changing speed demands, acceleration events, and road-grade profiles representative of diverse traffic and route conditions. These factors directly influence battery current draw and heat generation, leading to different thermal characteristics for Eco ($\kappa_{\text{agg}} = 0.25$), Normal ($\kappa_{\text{agg}} = 0.60$), and Aggressive ($\kappa_{\text{agg}} = 1.00$) driving styles. Simulated under open-loop cooling boundary $T_f = 45°\text{C}$ with constant flow resistance $R_u = 3.20\,\text{K/W}$ and initial state $T_c(0) = T_s(0) = 25°\text{C}$. The results demonstrate how road topology and driving behavior propagate through the electro-thermal system and determine the resulting battery temperature rise.

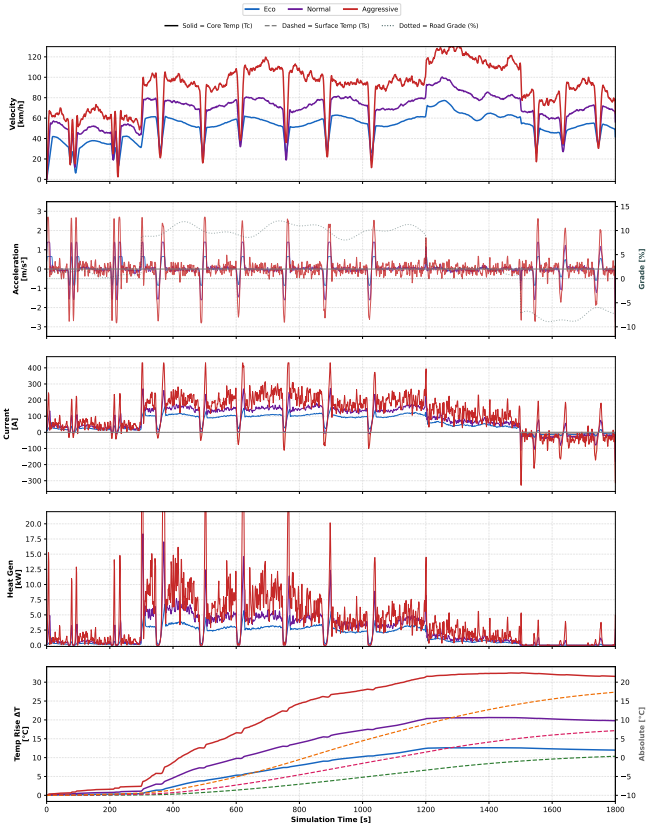


Fig. 2. Thermal and electrical response of the battery under eco, normal, and aggressive driving styles across varying road-grade profiles. Uphill segments and high-acceleration events increase battery current and heat generation, leading to greater temperature rise, while downhill operation and regenerative braking near the end of the drive cycle reduce current and thermal loading. Simulated under open-loop cooling boundary $T_f = 25°\text{C}$ with constant flow resistance $R_u = 3.20\,\text{K/W}$ and initial state $T_c(0) = T_s(0) = 25°\text{C}$. Due to battery thermal inertia, core temperature (T_c) increases faster and reaches higher values than surface temperature (T_s), with the largest thermal accumulation observed during aggressive driving despite transient reductions in load.

model incorporating SEI growth, capacity fade, and resistance propagation mechanisms.

Furthermore, a structured scenario-generation framework was developed to reproduce a wide range of operating conditions by combining environmental, vehicular, operational, and driver-context variables. The presented formulation establishes the foundation for predictive thermal management by enabling the estimation of battery heat generation, thermal evolution, and long-term degradation under realistic driving conditions. The generated dataset and simulation environment provide a reproducible benchmark for evaluating advanced control strategies and machine learning-based prediction algorithms.

The performance assessment of the proposed context-aware prediction architecture, supervisory decision layer and predictive thermal control strategy, together with comparisons against conventional reactive thermal management approaches, are presented in the accompanying conference paper. Future work will focus on controller validation through GT-SUITE co-simulation, real-time implementation, and the quantification of thermal-performance, energy-efficiency, and battery-lifetime trade-offs.

ACKNOWLEDGMENT

This research was supported by the Grant Agency of the Czech Technical University in Prague, grant No. SGS25/145/OHK3/3T/13

REFERENCES

[1] L. H. J. Raijmakers, D. L. Danilov, R.-A. Eichel, and P. H. L. Notten, "A review on various temperature-indication methods for Li-ion batteries," *Appl. Energy*, vol. 240, pp. 918–945, 2019, doi: 10.1016/j.apenergy.2019.03.008.

[2] A. Hales, L. Bravo Diaz, M. W. Marzook, Y. Zhao, Y. Patel, and G. Offer, "The cell cooling coefficient: A standard to define heat rejection from lithium-ion batteries," *J. Electrochem. Soc.*, vol. 166, no. 12, p. A2383, 2019, doi: 10.1149/2.0191912jes.

[3] A. Eldesoky *et al.*, "Impact of graphite materials on the lifetime of NMC811/graphite pouch cells: Part II. Long-term cycling, stack pressure growth, isothermal microcalorimetry, and lifetime projection," *J. Electrochem. Soc.*, vol. 169, no. 1, p. 010501, 2022, doi: 10.1149/1945-7111/ac42f1.

[4] T. Rahman and T. Alharbi, "Exploring lithium-ion battery degradation: A concise review of critical factors, impacts, data-driven degradation estimation techniques, and sustainable directions for energy storage systems," *Batteries*, vol. 10, no. 7, p. 220, 2024, doi: 10.3390/batteries10070220.

[5] M. S. Wahl, L. Spitthoff, H. I. Muri, A. Jinasena, O. S. Burheim, and J. J. Lamb, "The importance of optical fibres for internal temperature sensing in lithium-ion batteries during operation," *Energies*, vol. 14, no. 12, p. 3617, 2021, doi: 10.3390/en14123617.

[6] K. Kivekäs, J. Vepsäläinen, and K. Tammi, "Stochastic driving cycle synthesis for analyzing the energy consumption of a battery electric bus," *IEEE Access*, vol. 6, pp. 55586–55598, 2018, doi: 10.1109/ACCESS.2018.2871574.

[7] Q. Zhu, Y. Huang, C. F. Lee, P. Liu, J. Zhang, and T. Wik, "Predicting electric vehicle energy consumption from field data using machine learning," *IEEE Trans. Transport. Electrific.*, vol. 11, no. 1, pp. 2120–2132, Feb. 2025, doi: 10.1109/TTE.2024.3416532.

[8] J. Wang, I. Besselink, and H. Nijmeijer, "Battery electric vehicle energy consumption prediction for a trip based on route information," *Proc. Inst. Mech. Eng., D, J. Automob. Eng.*, vol. 232, no. 11, pp. 1528–1542, 2018, doi: 10.1177/0954407017729938.

[9] P. Rodríguez-Iturriaga, V. M. García, S. Rodríguez-Bolívar, E. E. Valdés, D. Anseán, and J. A. López-Villanueva, "A coupled electrothermal lithium-ion battery reduced-order model including heat generation due to solid diffusion," *Appl. Energy*, vol. 367, p. 123327, 2024, doi: 10.1016/j.apenergy.2024.123327.

[10] X. Lin, H. E. Perez, J. B. Siegel, A. G. Stefanopoulou, Y. Li, R. D. Anderson, Y. Ding, and M. P. Castanier, "Online parameterization of lumped thermal dynamics in cylindrical lithium ion batteries for core temperature estimation and health monitoring," *IEEE Trans. Control Syst. Technol.*, vol. 21, no. 5, pp. 1745–1755, Sep. 2013, doi: 10.1109/TCST.2012.2216267.

[11] H. Mediouni, A. Ezzouhri, Z. Charouh, K. El Harouri, S. El Hani, and M. Ghogho, "Energy consumption prediction and analysis for electric vehicles: A hybrid approach," *Energies*, vol. 15, no. 17, p. 6490, 2022, doi: 10.3390/en15176490.

[12] M. V. Morganti, S. Longo, M. Tirovic, C.-Y. Blaise, and G. Forostovsky, "Multi-scale, electro-thermal model of NMC battery cell," *IEEE Trans. Veh. Technol.*, vol. 68, no. 11, pp. 10594–10606, Nov. 2019, doi: 10.1109/TVT.2019.2943052.

[13] S. Kim, J. Jeong, J. Jin, I. Jung, and K.-Y. Oh, "Stochastic operational driving cycles of an electric vehicle battery system for degradation experiments," *Int. J. Energy Res.*, vol. 2026, p. 9134230, 2026, doi: 10.1155/er/9134230.